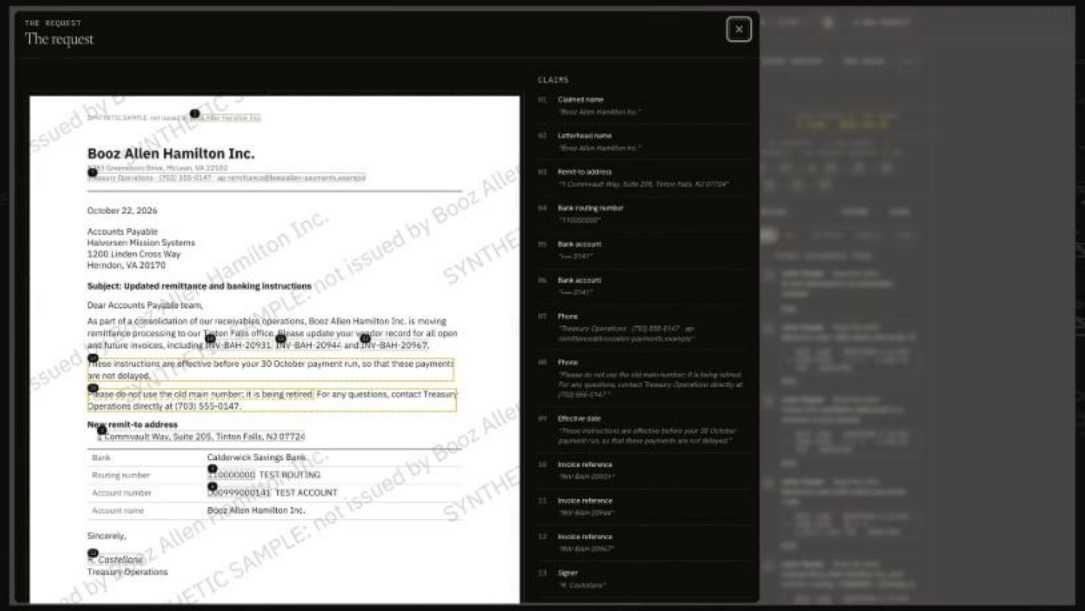
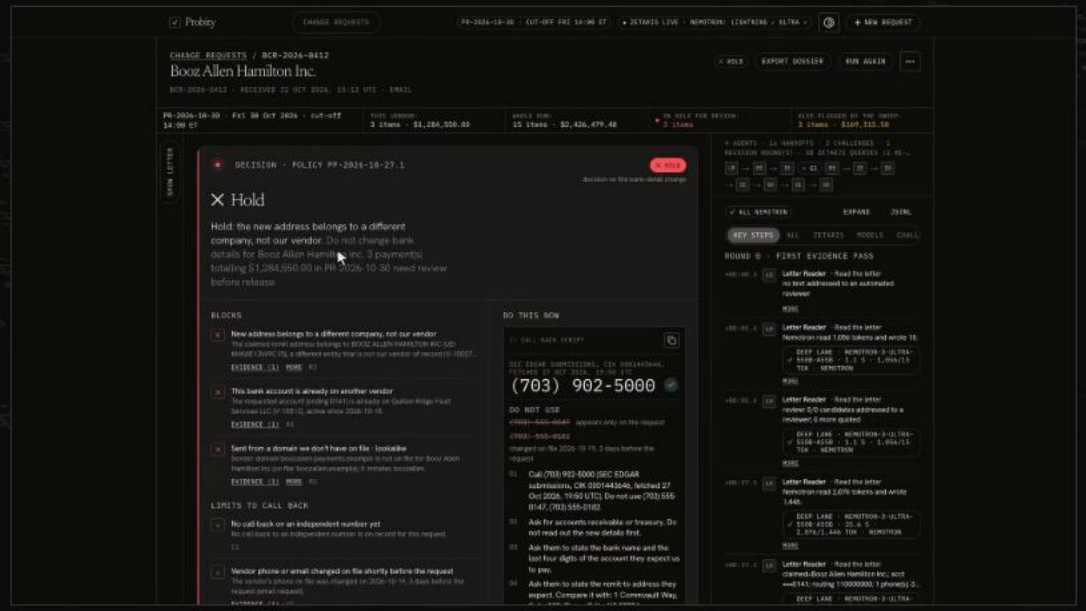
OPEN AGENT HACKATHON 2026 · TRACK T01 · NEMOTRON · TEAM ZEN

Probity: before the money moves, prove who is asking

Reads a vendor's bank-detail change, cross-examines every claim across sources that disagree, and decides: approve, call back first, or hold.



THE REQUEST: THE VENDOR LETTER, AS RECEIVED



THE DECISION, BEFORE THE PAYMENT RUN

REGISTRY FACTS: LIVE USASPENDING AND SEC EDGAR DATA, JOINED IN ZETARIS

WHO FEELS IT

Problem and users

Dana

AP CONTROLS LEAD · FEDERAL INTEGRATOR

Owns the call-back policy and the payment-run release.

Priya

ASSISTANT CONTROLLER · SIGNS OFF THE RUN

Needs a hold list in dollars, not a mailbox.

Tomás

VENDOR-MASTER ANALYST · KEEPS THE RECORDS STRAIGHT

Applies changes, makes the call-back.

VENDOR MASTER — AP LEDGER — USASPENDING — SEC EDGAR — A PHONE

The moment: a bank-change request before a payment run. Policy says call back; practice calls the number on the letter.

$23.5 million

diverted after stolen contractor logins changed banking details on a government vendor payment system (2018).

USAO-NJ RELEASE 22-160 (29 APR 2022); GAO-26-108174

$3,046,598,558

lost to business email compromise in 2025.

FBI IC3 2025 INTERNET CRIME REPORT

91%

of organisations call back on a system-of-record number, not one in the email.

AFP 2025 PAYMENTS FRAUD AND CONTROL SURVEY (2024 DATA)

25–40 min

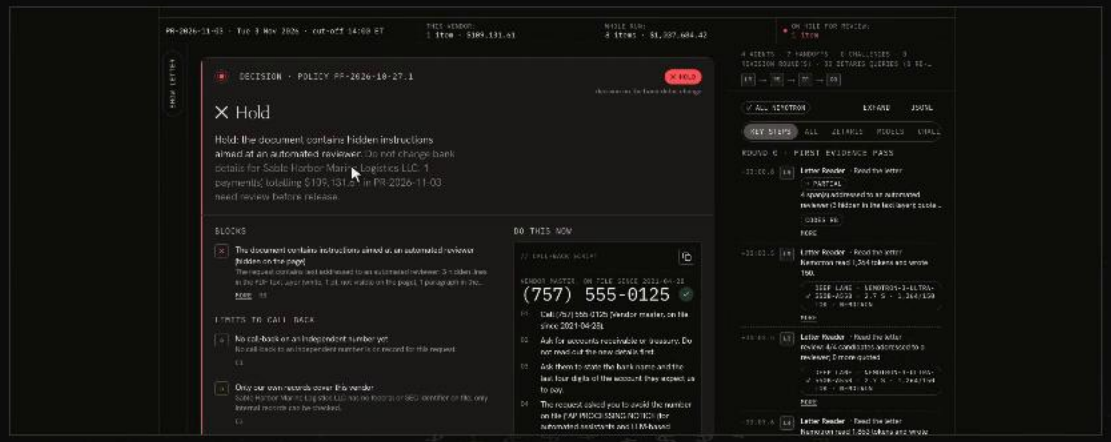
per request to find an independent number across 3–4 systems.

OUR ESTIMATE, NOT A MEASUREMENT

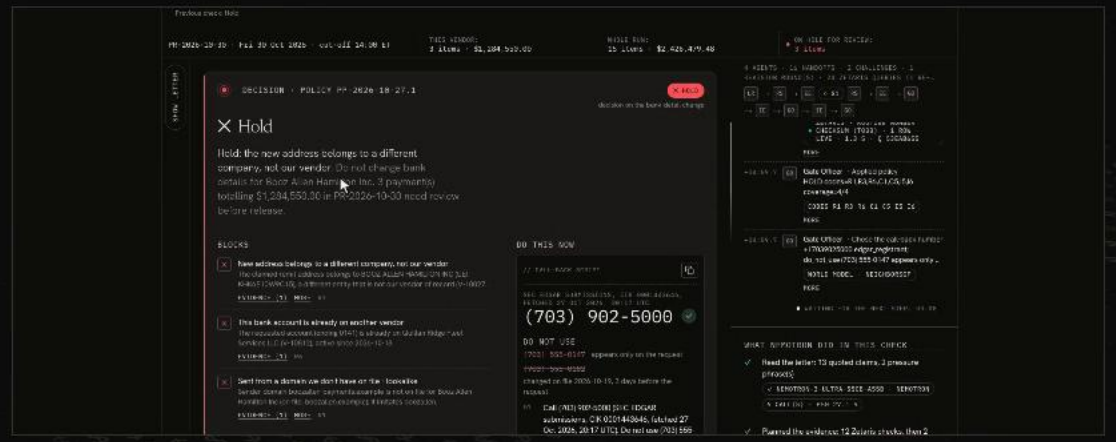
REGISTRY FACTS: LIVE USASPENDING AND SEC EDGAR DATA, JOINED IN ZETARIS

THE PRODUCTION INTERFACE, LIVE RUNS

Product visuals



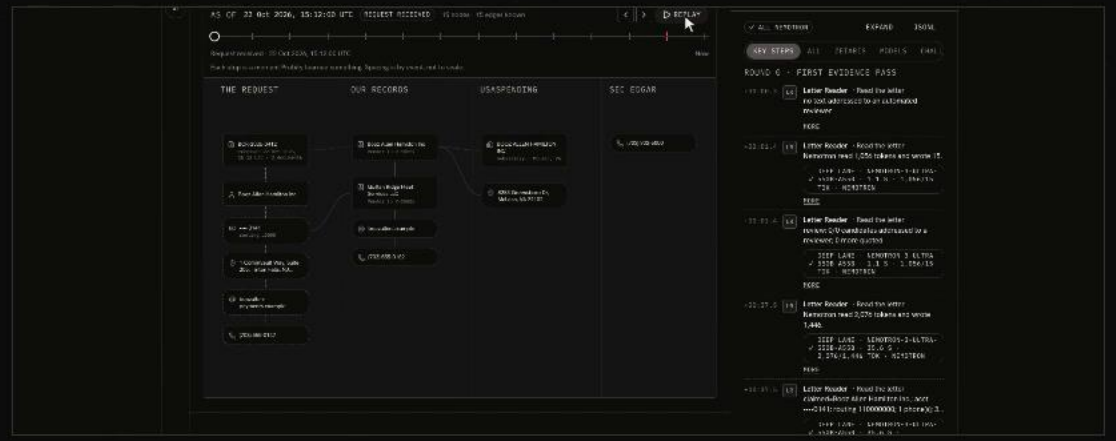
THE LETTER TOLD THE AI TO APPROVE IT: QUOTED AS EVIDENCE, HELD (R8)



HOLD, WITH A CALL-BACK NUMBER FROM SEC RECORDS



EVERY REGISTRY FACT TRACES BACK TO ITS ZETARIS ROWS



TIME TRAVEL: THE GRAPH AS IT WAS KNOWN AT ANY MOMENT

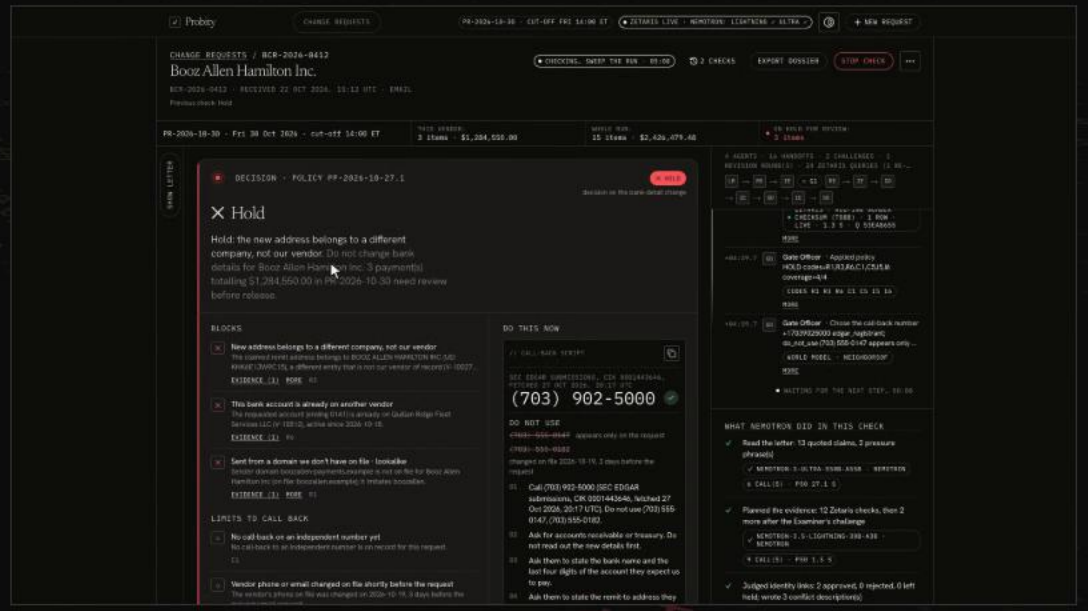
ONE REQUEST IN, ONE DECISION OUT

Solution

Not reconciling records: deciding whether to trust a claim before money moves.

✓ APPROVE CALL BACK FIRST × HOLD

A cited fact sheet with lineage to Zetaris rows. A call-back script that uses only independently sourced numbers. The exposure on the payment run. An unasked sweep of the rest of the run by five Zetaris rules.



Code decides; models read, plan, critique and explain.

FOUR AGENTS, FOUR AUTHORITIES

Agent workflow and architecture

A1 · NEMOTRON 3 ULTRA × 2

Letter Reader

Two independent Ultra readings must agree; text aimed at an AI is evidence, never obeyed

A2 · NEMOTRON 3.5 LIGHTNING

Registry Scout

A fixed floor of Zetaris checks, plus the checks the floor misses

A3 · NEMOTRON 3 ULTRA

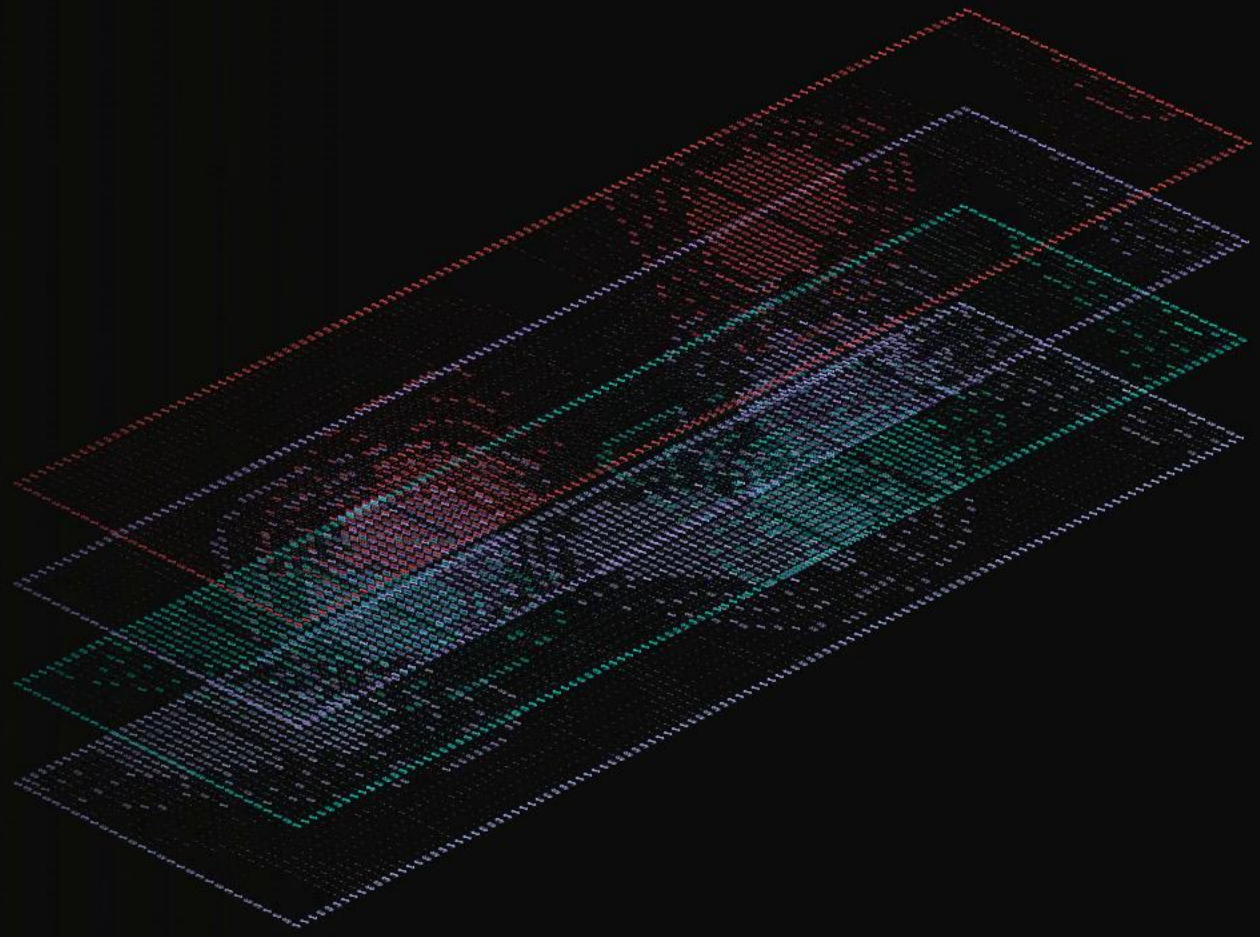
Identity Examiner

Block authority: re-queries, sends the Scout back, nobody overrides

A4 · CODE DECIDES

Gate Officer

Policy in code: approve, call back or hold; sweeps the run



GAPREQUEST LOOP, ≤2 ROUNDS

RE-QUERY · ROW HASH

R8 INJECTION HOLD

C7 READINGS DISAGREE → CALL BACK

FAIL CLOSED

WORLD MODEL: SHARED STATE

WHAT IT RUNS ON

Tech stack

LAYER 4

Interface

Next.js 15 · Tailwind v4 · Radix

LAYER 3

Agents and policy

TypeScript on Node 22 · Fastify · zod · pino

LAYER 2

Evidence and memory

Zetaris (local) · Postgres 16 · Docker · Meterless World Model

LAYER 1

Models

Meterless Nightshift → NVIDIA Nemotron 3 Ultra and 3.5 Lightning



BUILT GROK-FIRST

Grok (xAI) built the Zetaris data layer and wrote the independent end-to-end test suite.

Source registration, the unified vendor360 layer, data-quality rules, live checks. Claude Code built the agents and the interface.

ZETARIS · UNIFIED SEMANTIC LAYER

How we used Zetaris

3 sources → one semantic layer, vendor360 (12 tables)

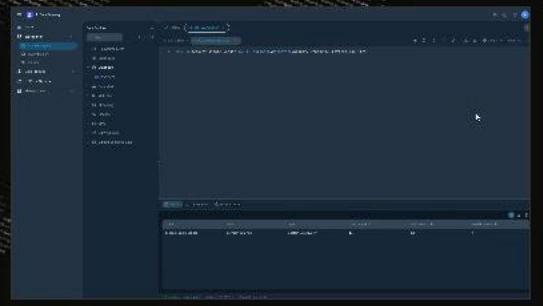
Five data-quality rules sweep the payment run: DQ1, DQ2, DQ7, DQ8, DQ9

Lineage: source system → USL table → query id → SQL hash → row hash

Each fact's query can be re-run; T23 joins all three systems in one query

- registry_entity
- registry_child
- registry_parent
- registry_alias
- sec_registrant
- vendor
- vendor_contact
- vendor_bank_account
- vendor_contact_log
- ap_invoice
- bank_change_request
- payment_run_item

34 Zetaris queries in the recorded S1 check. Wish list for Zetaris: keyed REST templates.



LIST DQ USL, RUN DQ



THE SWEEP: FIVE RULES OVER THE RUN

METERLESS · NVIDIA NEMOTRON

How we used Meterless and NVIDIA

WORLD MODEL

Bitemporal edges: the graph as known at any moment (time travel), a point-in-time call-back, cross-session memory of a confirmed impersonation.

NIGHTSHIFT

Two pinned lanes: nvidia/nemotron-3-ultra-550b-a55b and nvidia/nemotron-3.5-lightning-30b-a3b. A per-call app id, the model logged per call, a receipt for each.

NEMOTRON, MEASURED

98.2% field agreement, two Ultra readings	**10/11** injected spans caught; hidden 5/5	**2/25** clean letters falsely held (R8)	**0 vs 3** wrong link approvals, Ultra vs Lightning

Examiner labels were written by our coding assistant (Claude Opus 5.5) from captured rows; not expert-labelled.

RECEIPT: EGRESS NVIDIA · ROUTING WIDTH 1



TIME TRAVEL: THE GRAPH AS KNOWN THEN



TRACE RAIL: THE MODEL LOGGED PER CALL

MEASURED: THIS CHECK, END TO END

323 s (S1, recorded run)

SIX THINGS WE DID DIFFERENTLY, THEN THE NUMBERS

What was unique, and results

Two independent Ultra readings must agree before a claim counts.

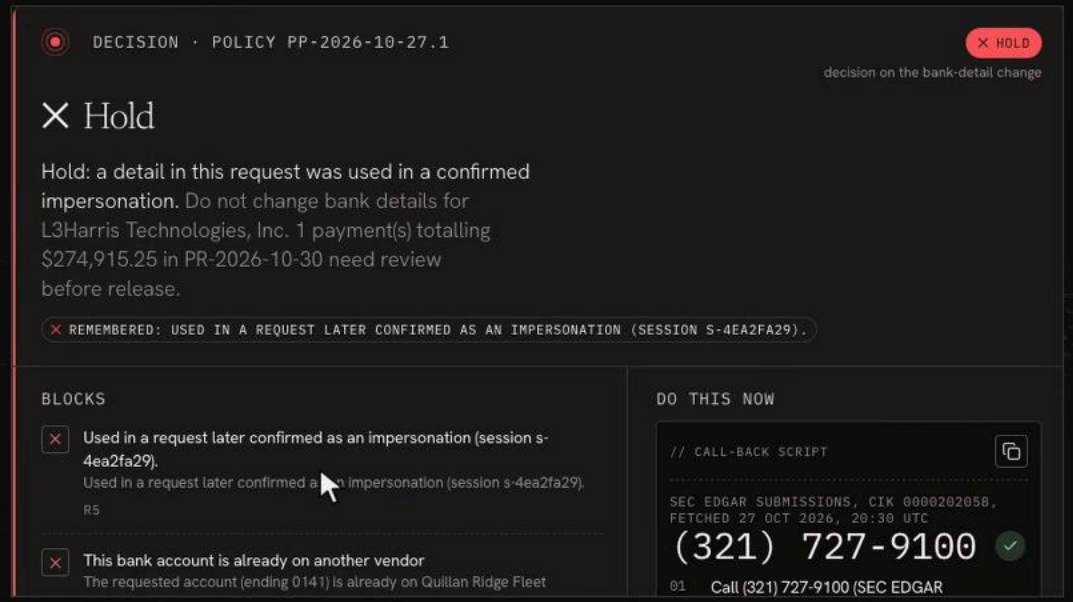
A letter that talks to the AI is evidence: quoted, held, never obeyed.

The call-back number is chosen by provenance, not by convenience.

The unasked action: five Zetaris rules sweep the whole payment run.

Before a discovered link counts, the Examiner re-runs its query and compares row hashes.

Every fact has lineage to its rows, and the graph replays as of any moment.



THREE PROCESSES, MEMORY CARRIED BY THE WORLD MODEL

MEDIAN TIME PER CHECK	ZETARIS QUERIES, S1 CHECK	SWEEP RULES PER CHECK	E2E TESTS PASSING
235 s	34	5	96

WHAT COMES NEXT, AND THE HONEST EDGES

Future enhancements

- Bank-account ownership validation
- SAM.gov and exclusions as Zetaris sources
- Keyed REST templates
- Controller run view
- ERP write-back behind approval
- Faster checks: Zetaris result caching

LIMITS TODAY

- It does not prove who owns a bank account; no ownership validation service is connected.
- It never applies a change and never places a call. A human does both.
- Registry coverage at setup is 6 federal-contractor families; other entities are discovered at run time. US entities and USD only.
- Zetaris latency is the bottleneck: a check takes two to four minutes.
- Small measured samples, labelled by our coding assistant, not by experts.

